



# Face Recognition-Based Attendance System

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# Project Overview

## Project Domain/Area

The project falls under the domain of Computer Vision.

## Project Description

The *Face Recognition-Based Attendance System* tracks attendance using facial recognition technology. Traditional methods are time-consuming and prone to errors. Attendees will look into a camera and press a button to mark their attendance, eliminating the need for manual roll calls. It addresses issues like time consumption and proxy attendance.

## Project Objectives

- Develop a real-time face detection module that accurately detects faces under varying lighting conditions.
- Implement a face recognition algorithm achieving at least 95% accuracy in identifying enrolled students/instructors.
- Design a user-friendly interface to enroll students/instructors, monitor real-time attendance, and generate reports.

## Target Audience/Users

The system targets instructors and administrative staff in educational institutions who need a reliable method to track student attendance. These users require a solution that saves time and provides accessible attendance records.

# Technical Approach

## AI Techniques

We plan to use Convolutional Neural Networks (CNNs) for facial feature extraction and recognition due to their good performance in image classification tasks. For real-time face detection and image processing, we will utilize OpenCV, which provides efficient tools for handling images and video streams. Additionally, we will use Haar Cascades for initial face detection due to their speed and effectiveness in identifying facial regions.

These techniques are appropriate due to their proven performance and open-source availability, ensuring both accuracy and development efficiency.

## Data Requirements

We will use the **Labeled Faces in the Wild (LFW)** dataset, which contains over 13,000 images of faces collected from the web, each labeled with the name of the person.

## System Architecture

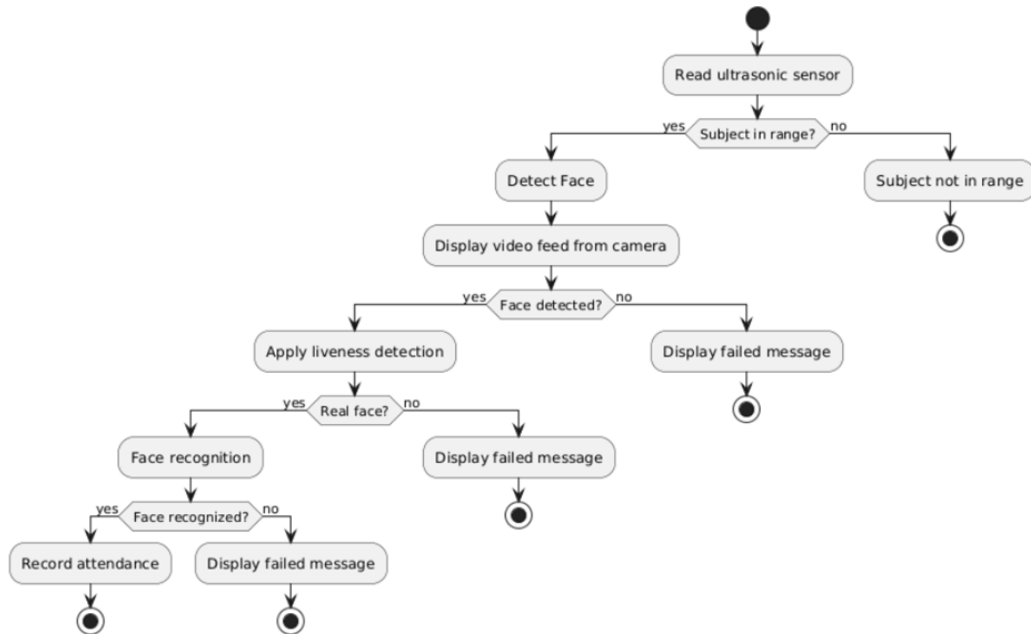


Figure 1: System Architecture Diagram

The system comprises:

- **Camera Input:** Captures live video feed.
- **Face Detection Module:** Detects faces using CNN.
- **Database:** Stores student profiles and attendance records.

# GUI Interface Design

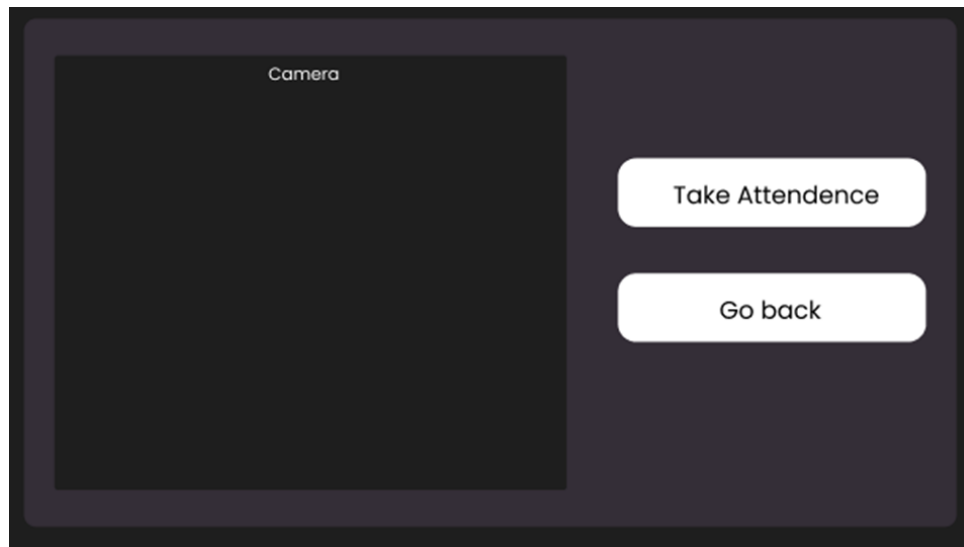


Figure 2: Camera Input Screen

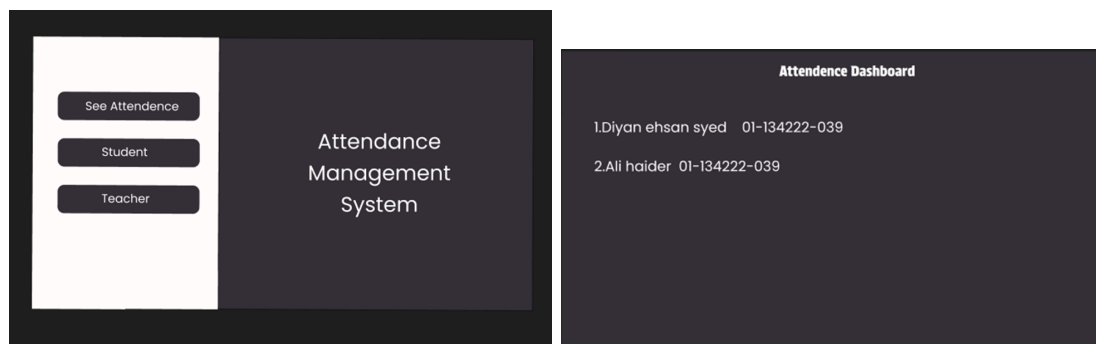


Figure 3: (Left) Home Screen (Right) Attendance Dashboard

Figma Link: [Click here](#)

## Resources and Tools

The project will use Python as the main programming language. We'll work with tools like OpenCV to handle images, TensorFlow and Keras to build the recognition model, and scikit-learn for classification. SQLite will serve as a lightweight database for storing student information and attendance records. Development will be done in Jupyter Notebook and Visual Studio Code, while LaTeX will be used to format the project report professionally.

## Team Roles and Responsibilities

We haven't finalized our roles yet.